

An **SQL DDL (Data Definition Language) script** is a text file filled with database commands used to actually **build, alter, or define the structural architecture** of a database.

While your ER diagram is a visual blueprint, the DDL script is the concrete code that tells your database engine (like MySQL or PostgreSQL) to create the physical tables, column data types, primary keys, and foreign key constraints.

The 3 Core DDL Commands You Use

- **CREATE:** Building brand-new database components (e.g., CREATE TABLE table_name (...)).
- **ALTER:** Modifying an existing structure (e.g., adding a column or adding a foreign key constraint).
- **DROP:** Obliterating a table or database completely to start fresh.

Part 1: Your Ready-to-Run DDL Script (Task 2 Setup)

Here is the exact **SQL DDL script** matching your corrected **draw.io** schema layout. Notice that the independent lookup tables are created first so that the dependent foreign key constraints don't break during execution:

```
-- 1. Create Patient Table (Independent)
CREATE TABLE Patient (
    Patient_ID INT AUTO_INCREMENT,
    Name VARCHAR(100) NOT NULL,
    DoB DATE NOT NULL,
    Gender VARCHAR(20),
    Address VARCHAR(255),
    Telephone_Number VARCHAR(20),
    Emergency_Contact_Info VARCHAR(255),
    Life_Status VARCHAR(50),
    CONSTRAINT PK_Patient PRIMARY KEY (Patient_ID)
);

-- 2. Create ICD9 Dictionary Table (Independent Reference)
CREATE TABLE ICD9_Dictionary (
    ICD9_Code VARCHAR(20),
    Official_Diagnosis_Title VARCHAR(255) NOT NULL,
    Description TEXT,
    Disease_Category VARCHAR(100),
    CONSTRAINT PK_ICD9 PRIMARY KEY (ICD9_Code)
);
```

```

-- 3. Create Core Admission Table (Depends on Patient)
CREATE TABLE Admission (
    Admission_ID INT AUTO_INCREMENT,
    Patient_ID INT NOT NULL,
    Admission_Date_Time DATETIME NOT NULL,
    Admission_Type VARCHAR(50),
    Reason_for_Admission VARCHAR(255),
    Admission_Source VARCHAR(100),
    Insurance_Information VARCHAR(255),
    Admitting_Professional_Name VARCHAR(100),
    Final_Admission_Status VARCHAR(50),
    CONSTRAINT PK_Admission PRIMARY KEY (Admission_ID),
    CONSTRAINT FK_Admission_Patient FOREIGN KEY (Patient_ID)
        REFERENCES Patient(Patient_ID) ON DELETE CASCADE
);

-- 4. Create Triage Assessment Table (Depends on Admission)
CREATE TABLE Triage_Assessment (
    Triage_ID INT AUTO_INCREMENT,
    Admission_ID INT NOT NULL,
    Blood_Pressure VARCHAR(20),
    Heart_Rate INT,
    Respiratory_Rate INT,
    Body_Temperature DECIMAL(4,2),
    Oxygen_Saturation INT,
    Pain_Level INT,
    Reported_Symptoms TEXT,
    Known_Allergies TEXT,
    Current_Medications TEXT,
    Triage_Priority_Level VARCHAR(50),
    CONSTRAINT PK_Triage PRIMARY KEY (Triage_ID),
    CONSTRAINT FK_Triage_Admission FOREIGN KEY (Admission_ID)
        REFERENCES Admission(Admission_ID) ON DELETE CASCADE
);

-- 5. Create Clinical Note Table (Depends on Admission)
CREATE TABLE Clinical_Note (
    Note_ID INT AUTO_INCREMENT,
    Admission_ID INT NOT NULL,
    Authoring_Physician VARCHAR(100),
    Note_Type VARCHAR(50),
    Creation_Date_Time DATETIME NOT NULL,
    Contents TEXT,
    CONSTRAINT PK_Clinical_Note PRIMARY KEY (Note_ID),
    CONSTRAINT FK_Note_Admission FOREIGN KEY (Admission_ID)
        REFERENCES Admission(Admission_ID) ON DELETE CASCADE
);

```

-- 6. Create Diagnostic Order Table (Depends on Admission and Patient)

```
CREATE TABLE Diagnostic_Order (  
    Order_ID INT AUTO_INCREMENT,  
    Admission_ID INT NOT NULL,  
    Patient_ID INT NOT NULL,  
    Requesting_Physician VARCHAR(100),  
    Requested_Examination VARCHAR(100),  
    Request_Date_Time DATETIME NOT NULL,  
    Clinical_Reason VARCHAR(255),  
    Priority VARCHAR(50),  
    Order_Status VARCHAR(50),  
    CONSTRAINT PK_Diagnostic_Order PRIMARY KEY (Order_ID),  
    CONSTRAINT FK_Order_Admission FOREIGN KEY (Admission_ID)  
        REFERENCES Admission(Admission_ID) ON DELETE CASCADE,  
    CONSTRAINT FK_Order_Patient FOREIGN KEY (Patient_ID)  
        REFERENCES Patient(Patient_ID) ON DELETE CASCADE  
);
```

-- 7. Create Radiology Examination Table (Depends on Admission)

```
CREATE TABLE Radiology_Examination (  
    Examination_ID INT AUTO_INCREMENT,  
    Admission_ID INT NOT NULL,  
    Examination_Type VARCHAR(100),  
    Examination_Date_Time DATETIME NOT NULL,  
    Body_Area VARCHAR(100),  
    Referring_Physician VARCHAR(100),  
    Radiology_Technician VARCHAR(100),  
    Examination_Status VARCHAR(50),  
    Image_Link VARCHAR(255),  
    Radiologist_Findings TEXT,  
    CONSTRAINT PK_Radiology PRIMARY KEY (Examination_ID),  
    CONSTRAINT FK_Radiology_Admission FOREIGN KEY (Admission_ID)  
        REFERENCES Admission(Admission_ID) ON DELETE CASCADE  
);
```

-- 8. Create Surgical Procedure Table (Depends on Admission)

```
CREATE TABLE Surgical_Procedure (  
    Procedure_ID INT AUTO_INCREMENT,  
    Admission_ID INT NOT NULL,  
    Procedure_Type VARCHAR(100),  
    Procedure_Date_Time DATETIME NOT NULL,  
    Surgeon_Name VARCHAR(100),  
    CONSTRAINT PK_Surgery PRIMARY KEY (Procedure_ID),  
    CONSTRAINT FK_Surgery_Admission FOREIGN KEY (Admission_ID)  
        REFERENCES Admission(Admission_ID) ON DELETE CASCADE  
);
```

```

-- 9. Create Unit Stay Table (Depends on Admission)
CREATE TABLE Unit_Stay (
    Unit_Stay_ID INT AUTO_INCREMENT,
    Admission_ID INT NOT NULL,
    Unit_Type VARCHAR(50),
    Ward_Room_Number VARCHAR(20),
    Bed_Number VARCHAR(20),
    Entry_Date_Time DATETIME NOT NULL,
    Exit_Date_Time DATETIME,
    Responsible_Physician VARCHAR(100),
    Reason_for_Transfer VARCHAR(255),
    CONSTRAINT PK_Unit_Stay PRIMARY KEY (Unit_Stay_ID),
    CONSTRAINT FK_Unit_Stay_Admission FOREIGN KEY (Admission_ID)
        REFERENCES Admission(Admission_ID) ON DELETE CASCADE
);

-- 10. Create Diagnosis Bridge Table (Many-to-Many Bridge)
CREATE TABLE Diagnosis (
    Diagnosis_ID INT AUTO_INCREMENT,
    Admission_ID INT NOT NULL,
    Patient_ID INT NOT NULL,
    ICD9_Code VARCHAR(20) NOT NULL,
    Diagnosis_Date_Time DATETIME NOT NULL,
    Diagnosis_Type VARCHAR(50),
    Diagnosing_Physician VARCHAR(100),
    CONSTRAINT PK_Diagnosis PRIMARY KEY (Diagnosis_ID),
    CONSTRAINT FK_Diag_Admission FOREIGN KEY (Admission_ID)
        REFERENCES Admission(Admission_ID) ON DELETE CASCADE,
    CONSTRAINT FK_Diag_Patient FOREIGN KEY (Patient_ID)
        REFERENCES Patient(Patient_ID) ON DELETE CASCADE,
    CONSTRAINT FK_Diag_ICD9 FOREIGN KEY (ICD9_Code)
        REFERENCES ICD9_Dictionary(ICD9_Code) ON DELETE RESTRICT
);

-- 11. Create Discharge Record Table (1-to-1 with Admission)
CREATE TABLE Discharge_Record (
    Discharge_ID INT AUTO_INCREMENT,
    Admission_ID INT NOT NULL UNIQUE, -- UNIQUE guarantees a 1-to-1
relationship limit
    Discharge_Date_Time DATETIME NOT NULL,
    Discharge_Destination VARCHAR(100),
    Condition_at_Discharge VARCHAR(100),
    Prescribed_Medications TEXT,
    FollowUp_Instructions TEXT,
    Responsible_Physician VARCHAR(100),
    CONSTRAINT PK_Discharge PRIMARY KEY (Discharge_ID),
    CONSTRAINT FK_Discharge_Admission FOREIGN KEY (Admission_ID)
        REFERENCES Admission(Admission_ID) ON DELETE CASCADE
);

```

) ;

Would you like me to walk you through how to build the **SQL DML (Data Manipulation Language) insert scripts** next so you can populate these clean tables with fake patient dummy records, or would you prefer a list of **common tricky questions** professors use to test your schema design knowledge during your presentation defense?